

CODE-PART-1

```
1 #include <stdio.h>
2 int midtermAverage , finalAverage;
3 char first_initial(FILE *id_fp , int id) {
4     char karakter;
5     int okunan_sayi = 0;
6
7     while ((karakter = fgetc(id_fp)) != EOF) {
8
9         while (fscanf(id_fp, "%d;", &okunan_sayi) == 1) {
10             if(okunan_sayi == id){
11                 karakter = fgetc(id_fp);
12             }
13         }
14     }
15     return karakter;
16 }
17
18 char last_initial(FILE *id_fp , int id) {
19     char karakter;
20     int okunan_sayi = 0 , okunan_noktali_virgul = 0;
21
22     while ((karakter = fgetc(id_fp)) != EOF) {
23
24         while (fscanf(id_fp, "%d;", &okunan_sayi) == 1) {
25             if(okunan_sayi == id){
26
27                 while((karakter = fgetc(id_fp)) != EOF && okunan_noktali_virgul < 1 ){
28
29                     if(karakter == ';'){
30                         okunan_noktali_virgul++;
31                     }
32                     if(okunan_noktali_virgul > 0){ /* adin sonundaki noktali virgul geldikten sonra*/
33                         karakter = fgetc(id_fp);
34                     }
35                 }
36             }
37         }
38     }
39     return karakter;
40 }
```

In this part; in first function , I read the numbers one by one. If the number read is equal to the entered ID, I took the next letter.(First letter of name.)

In second function, I did similar things. But if the number read is equal to ID, I took the letter after the second semicolon. (First letter of surname.)

CODE-PART-2

```
41
42 void calculate_average_midterm(FILE *id_fp ){
43     int okunan_sayi = 0;
44     int student_id,not1,not2,not3,not4;
45     int total_midterms = 0 , number_of_students = 0 , average_of_midterms = 0;
46
47     /* average midterm */
48     while (fscanf(id_fp, "%d;%d;%d;%d;%d\n", &student_id, &not1, &not2, &not3, &not4) == 5) {
49
50         int average_of_midterms = (not1 + not2) / 2;
51         total_midterms += average_of_midterms;
52         number_of_students++;
53     }
54     while(fscanf(id_fp, "%d;%d;%d;%d\n", &student_id, &not1, &not2, &not3) == 4) {
55         total_midterms += not1;
56         number_of_students++;
57     }
58     average_of_midterms = (total_midterms / number_of_students);
59     midtermAverage = average_of_midterms;
60
61 }
62 }
63 void calculate_average_final(FILE *id_fp ){
64     int okunan_sayi = 0;
65     int student_id,not1,not2,not3,not4;
66     int total_final = 0 , number_of_students = 0 , average_of_final = 0;
67
68     /* average final */
69     while (fscanf(id_fp, "%d;%d;%d;%d;%d\n", &student_id, &not1, &not2, &not3, &not4) == 5) {
70
71         total_final += not3;
72         number_of_students++;
73     }
74     while(fscanf(id_fp, "%d;%d;%d;%d\n", &student_id, &not1, &not2, &not3) == 4) {
75         total_final += not1;
76         number_of_students++;
77     }
78     average_of_final = (total_final / number_of_students);
79     finalAverage = average_of_final;
80
81 }
```

In this part, I found the average midterm and final values and set them equal to the global variables I defined at the beginning.

CODE-PART-3

```
83 int average_grade(FILE *id_fp , int id) {
84     char karakter;
85     int okunan_sayi = 0;
86     int student_id,not1,not2,not3,not4;
87     int GPA = 0;
88     /* return gpa*/
89     while (fscanf(id_fp, "%d;%d;%d;%d;%d\n", &student_id, &not1, &not2, &not3, &not4) == 5) {
90
91         int midterm = (not1+not2);
92         if(student_id == id){
93             if(midterm < 40 && midterm < midtermAverage && (not3 < 40 && not3 < finalAverage)){
94                 GPA = 1;
95             }else if(midterm > 40 && midterm < midtermAverage && (not3 > 40 && not3 < finalAverage)){
96                 GPA = 2.5;
97             }else if(midterm > 40 && midterm < midtermAverage && (not3 > 40 && not3 > finalAverage)){
98                 GPA = 3;
99             }else if(midterm > 40 && midterm > midtermAverage && (not3 < 90 && not3 > finalAverage)){
100                 GPA = 3.5;
101             }else if(midterm > 90 && (not3 < 90)){
102                 GPA = 4;
103             }
104         }
105     }
106
107     while(fscanf(id_fp, "%d;%d;%d;%d\n", &student_id, &not1, &not2, &not3) == 4) {
108
109         int midterm = not1;
110         if(student_id == id){
111             if(midterm < 40 && midterm < midtermAverage && (not2 < 40 && not2 < finalAverage)){
112                 GPA = 1;
113             }else if(midterm > 40 && midterm < midtermAverage && (not2 > 40 && not2 < finalAverage)){
114                 GPA = 2.5;
115             }else if(midterm > 40 && midterm < midtermAverage && (not2 > 40 && not2 > finalAverage)){
116                 GPA = 3;
117             }else if(midterm > 40 && midterm > midtermAverage && (not2 < 90 && not2 > finalAverage)){
118                 GPA = 3.5;
119             }else if(midterm > 90 && (not2 < 90)){
120                 GPA = 4;
121             }
122         }
123     }
124     printf("Student's GPA is: %d \n" , GPA);
125     return GPA;
126 }
```

In this part, If there were 5 numbers in the line read, I took the average of the midterms. Then I compared the values I found with global variables and calculated the GPA. I determine the line read according to the ID I receive from the user. The ID entered by the user represents studentID.

CODE-PART-4

```
128 int get_id_fi(FILE * id_fp, char first_initial){
129     FILE *students1;
130     students1 = fopen("//home//gorkem//Desktop//students1.txt", "w");
131
132     char karakter;
133     int okunan_sayi = 0;
134
135     printf("List of students whose names start with %c: \n", first_initial);
136     while ((karakter = fgetc(id_fp)) != EOF) {
137         while (fscanf(id_fp, "%d;", &okunan_sayi) == 1) {
138             karakter = fgetc(id_fp);
139             if(karakter == first_initial){
140                 fprintf(students1, "%d\n", okunan_sayi); /* birden fazla ogrenciyi return edemedigim icin bu sekilde printledim.*/
141             }
142         }
143     }
144     fclose(students1);
145     /* dosya okuna */
146     FILE *students1_1;
147     students1_1 = fopen("//home//gorkem//Desktop//students1.txt", "r");
148     if (students1_1 == NULL) {
149         printf("Dosya acilamadi.\n");
150         return 1;
151     }
152     int isEmpty = fgetc(students1_1);
153     fclose(students1_1);
154     students1_1 = fopen("//home//gorkem//Desktop//students1.txt", "r"); /* acip kapatmamin sebebi bos olup olmadigini kontrol etmek.*/
155     char ch;
156     if (isEmpty == EOF) {
157         printf("There is no result.\n");
158     }else{
159         while ((ch = fgetc(students1_1)) != EOF) {
160             printf("%c", ch);
161         }
162     }
163     fclose(students1_1);
164     return 0;
165 }
```

In this part; I added the students whose names started with the letter I received from the user to a separate file and read that file.

CODE-PART-5

```
100
167 int get_id_li(FILE *id_fp, char last_initial) {
168
169     FILE *students2;
170     students2 = fopen("//home//gorkem//Desktop//students2.txt", "w");
171
172     char karakter;
173     int okunan_sayi = 0;
174
175     printf("List of students whose last-name start with %c: \n", last_initial);
176     while ((karakter = fgetc(id_fp)) != EOF) {
177         while (fscanf(id_fp, "%d;", &okunan_sayi) == 1) {
178
179             while(karakter != ';'){ /* soyismin oncesindeki noktali virgule kadar gel */
180                 karakter = fgetc(id_fp);
181
182             } /* soyismin ilk harfini oku*/
183             karakter = fgetc(id_fp);
184             if(karakter == last_initial){
185                 fprintf(students2, "%d\n", okunan_sayi);
186             }
187         }
188     }
189     fclose(students2);
190     /* dosya okuma */
191     FILE *students1_2;
192     students1_2 = fopen("//home//gorkem//Desktop//students2.txt", "r");
193     if (students1_2 == NULL) {
194         printf("Dosya acilamadi.\n");
195         return 1;
196     }
197     int isEmpty = fgetc(students1_2);
198     fclose(students1_2);
199     students1_2 = fopen("//home//gorkem//Desktop//students2.txt", "r");
200     char ch;
201     if (isEmpty == EOF) {
202         printf("There is no result.\n");
203     }else{
204         while ((ch = fgetc(students1_2)) != EOF) {
205             printf("%c", ch);
206         }
207     }
208     fclose(students1_2);
209     return 0;
210 }
```

In this part; I did same thing with part4. But I read up to the semicolon before the last name and then took the character after that semicolon. (First letter of last name.)

CODE-PART-6

```
213 void number_of_classes(FILE *id_fp, int user_id){
214     int id,classId,department,not3,not4;
215
216     while (fscanf(id_fp, "%d;%d;%d;%d;%d\n", &id, &classId, &department, &not3, &not4) == 5) {
217
218     }
219     while(fscanf(id_fp, "%d;%d;%d;%d\n", &id, &classId, &department, &not3) == 4) {
220
221     }
222     while(fscanf(id_fp, "%d;%d;%d\n", &id, &classId, &department) == 3) {
223         if(user_id == id){
224             printf("# of the classes for instructor: %d \n" ,classId );
225         }
226     }
227 }
228 void print_course_id(FILE *id_fp, int user_id){
229
230     int id,classId,department,not3,not4;
231     while (fscanf(id_fp, "%d;%d;%d;%d;%d\n", &id, &classId, &department, &not3, &not4) == 5) {
232         if(user_id == id){
233             printf("Course ID is : %d \n" ,not4 );
234         }
235     }
236     while(fscanf(id_fp, "%d;%d;%d;%d\n", &id, &classId, &department, &not3) == 4) {
237         if(user_id == id){
238             printf("Course ID is : %d \n" ,not3 );
239         }
240     }
241 }
242
243 void print_average_grade_of_course(FILE *id_fp, int course_id){
244     int id,midterm1,midterm2,final_n,courseId , totalGPA = 0 , totalStudent = 0;
245
246     while (fscanf(id_fp, "%d;%d;%d;%d;%d\n", &id, &midterm1, &midterm2, &final_n, &courseId) == 5) {
247         totalGPA += average_grade(id_fp, id);
248         totalStudent++;
249     }
250     while(fscanf(id_fp, "%d;%d;%d;%d\n", &id, &midterm1, &final_n, &courseId) == 4) {
251         totalGPA += average_grade(id_fp, id);
252         totalStudent++;
253     }
254     int averageGPA = totalGPA / totalStudent;
255     printf("Average GPA for class %d : %d" ,course_id , averageGPA );
256 }
257
```

In this part; in first function , I read the lines with 5 and 4 numbers using while loops just to make a comparison with the instructors' IDs. So I came to the lines, which are 3 numbers. (Instructors' line.)

In the second function, I got the course id according to the user id I received from the user.

In the third function, I found the average GPA value in that class according to the course ID I received from the user.

CODE-PART-7

```
258 int main() {
259     FILE *dosya;
260     FILE *dosya2;
261     char option;
262     int isCont = 1;
263     dosya = fopen("//home//gorkem//Desktop//first 1.txt", "r"); /* okuma şeklinde aç*/
264
265     dosya2 = fopen("//home//gorkem//Desktop//second 1.txt", "r"); /* okuma şeklinde aç*/
266
267     if (dosya == NULL) {
268         printf("Dosya bulunamadi.\n");
269         return 1;
270     }
271     if (dosya2 == NULL) {
272         printf("Dosya bulunamadi.\n");
273         return 1;
274     }
275     /* average final ve midterm i bulup global degiskeni degistirdim*/
276     calculate_average_midterm(dosya);
277     fclose(dosya);
278     dosya = fopen("//home//gorkem//Desktop//first 1.txt", "r"); /*okuma şeklinde aç*/
279     calculate_average_final(dosya);
280     fclose(dosya);
281 }
```

In this part; I opened files. Then I called the `calculate_average_midterm` and `calculate_average_final` functions. So I set the global variables with that values in this functions.

CODE-PART-8

```
281
282     while(isCont == 1){
283         char letterGrade;
284         int studentId , instructorId , userId , courseId;
285         printf("\nChoose an option: ");
286         fflush(stdin); /* Tamponu temizle */
287         scanf("%c" , &option);
288
289         if(option == 'p'){ /* yapamadim */
290             break;
291         }else if(option == 'n'){
292
293             char letter;
294             printf("Enter a letter for the initial:");
295             fflush(stdin); /* Tamponu temizle */
296             scanf("%c" , &letter);
297             dosya = fopen("//home//gorkem//Desktop//second 1.txt", "r");
298             get_id_fi(dosya,letter);
299             fclose(dosya);
300         }else if(option == 'g'){
301             printf("Enter the studentId for the GPA:");
302             scanf("%d" , &studentId);
303             dosya = fopen("//home//gorkem//Desktop//first 1.txt", "r");
304             int gpaOfStudent = average_grade(dosya ,studentId);
305             fclose(dosya);
306             if(gpaOfStudent == 1){
307                 letterGrade = 'F';
308                 printf("Studen'ts letter grade is:%c \n" , letterGrade);
309             }else if(gpaOfStudent == 2.5){
310                 letterGrade = 'D';
311                 printf("Studen'ts letter grade is:%c \n" , letterGrade);
312             }else if(gpaOfStudent == 3){
313                 letterGrade = 'C';
314                 printf("Studen'ts letter grade is:%c \n" , letterGrade);
315             }else if(gpaOfStudent == 3.5){
316                 letterGrade = 'B';
317                 printf("Studen'ts letter grade is:%c \n" , letterGrade);
318             }else if(gpaOfStudent == 4){
319                 letterGrade = 'A';
320                 printf("Studen'ts letter grade is:%c \n" , letterGrade);
321             }
322         }
```

In this part I created the menu.

CODE-PART-9

```
322 |  
323 |  
324 |     }else if(option == 'c'){ /* dogru calismiyor hatayi bulamadim*/  
325 |         printf("Enter the course ID:");  
326 |         scanf("%d" , &courseId);  
327 |         dosya = fopen("//home//gorkem//Desktop//first 1.txt", "r");  
328 |         print_average_grade_of_course(dosya , courseId);  
329 |         fclose(dosya);  
330 |     }else if(option == 't'){  
331 |         printf("Enter the instructor ID:");  
332 |         scanf("%d" , &instructorId);  
333 |         dosya = fopen("//home//gorkem//Desktop//first 1.txt", "r");  
334 |         number_of_classes(dosya , instructorId);  
335 |         fclose(dosya);  
336 |     }else if(option == 'd'){ /* yapamadim */  
337 |         break;  
338 |     }else if(option == 'l'){  
339 |         printf("Enter the USER ID:");  
340 |         scanf("%d" , &userId);  
341 |         if(userId < 200000000){  
342 |             printf("This user is not a student! \n");  
343 |         }else{  
344 |             dosya = fopen("//home//gorkem//Desktop//first 1.txt", "r"); /* okuma şeklinde aç */  
345 |             print_course_id(dosya , userId);  
346 |             fclose(dosya);  
347 |         }  
348 |     }else if(option == 'e'){  
349 |         printf("Exiting... \n");  
350 |         isCont = 0;  
351 |         break;  
352 |     }  
353 | }  
354 |  
355 | return 0;  
356 | }
```

Continue to menu.

OUTPUTS:

-UBUNTU-

```
Choose an option: g
Enter the studentId for the GPA:220015014
Student's GPA is: 3
Student's letter grade is:C
```

```
Choose an option:
Choose an option: c
Enter the course ID:108
Student's GPA is: 0
Average GPA for class 108 : 0
Choose an option:
Choose an option: l
Enter the USER ID:210015015
Course ID is : 108
```

```
Choose an option:
Choose an option: l
Enter the USER ID:195001003
This user is not a student!
```

```
Choose an option:
Choose an option: e
Exiting...
```

```
gorkem@gorkem:~/Desktop$
```

-windows-

```
Choose an option: n
Enter a letter for the initial:r
List of students whose names start with r:
220015022
210015002
210015012
210015024
195001005
195001009

Choose an option: g
Enter the studentId for the GPA:220015015
Student's GPA is: 3
Student's letter grade is:C

Choose an option: l
Enter the USER ID:220015003
Course ID is : 102

Choose an option: l
Enter the USER ID:210015013
Course ID is : 108

Choose an option: l
Enter the USER ID:195001009
This user is not a student!

Choose an option: e
Exiting...

Process returned 0 (0x0)    execution time : 73.530 s
Press any key to continue.
```

Dear teacher; on the ubuntu since the fflush function did not work , I could not clear the buffer and because of that the code did not work properly. But the same code worked without and problems on Windows.

YOUTUBE LINK: <https://youtu.be/fDu3nlyHINE>

Video Hocam yurtta kaldığım için odada ses çıktı ve videonun bazı yerlerini kırmak zorunda kaldım fakat videodan da anlayacağınız üzere kodda herhangi bir değişiklik yapmadan kaldığım yerden devam ettirdim.